Quizrr Chapter-Wise Test for JEE Advanced - 2025

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Q.1

The number of integral solutions of $|D| = 8$, where $D = \begin{vmatrix} y+z & z & y \\ z & z+x & x \\ y & x & x+y \end{vmatrix}$ is -

(A) 3

(B) 8

(C) 16

(D) 24

Q.2

If $\mathrm{f}(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$ and if α, β, γ, are angle of a triangle, then $\mathrm{f}(\alpha).\mathrm{f}(\beta).\mathrm{f}(\gamma)$ equals

(A) I

(B) -I

(C) 0

(D) None of these

Q.3

If a determinant of order 3×3 is formed by using the numbers of 1 or -1 then minimum value of determinant is -

(A) -2

(B) -4

(C) 0

(D) -8

Q.4

If $f(x) = \begin{vmatrix} x^n & \sin x & \cos x \\ n! & \sin\left(\frac{n\pi}{2}\right) & \cos\left(\frac{n\pi}{2}\right) \\ a & a^2 & a^3 \end{vmatrix}$, then find $\frac{d^n}{dx^n}[f(x)]$ at $x = 0$.

(A) 2

(B) 4

(C) 0

(D) 8

Q.5

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If $A = \begin{bmatrix} 0 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 0 \end{bmatrix}$ and I is identity matrix of order 2, then -

(A) $I + A = (I - A)\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$

(B) $I - A = (I + A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$

(C) $I + A = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

(D) None of these

Q.6

If $A = \begin{bmatrix} \mathrm{i} & 0 \\ 0 & \mathrm{i} \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -\mathrm{i} \\ -\mathrm{i} & 0 \end{bmatrix}$, then $(A+B)(A-B)$ is -

(A) $A^2 - B^2$

(B) $A^2 + B^2$

(C) $A^2 - B^2 + BA + AB$

(D) None of these

Q.7

Let $\mathrm{A} = \begin{bmatrix} 5\mathrm{a}^2 + 2\mathrm{bc} & 6 & 8 \\ 13 & 8\,\mathrm{b}^2 - 10\mathrm{ac} & -9 \\ -7 & 5 & 25\mathrm{c}^2 \end{bmatrix}, \mathrm{B} = \begin{bmatrix} \mathrm{a}^2 + 6\mathrm{ab} & 3 & 5 \\ 12 & -\mathrm{b}^2 & 6 \\ 1 & 4 & 17\mathrm{bc} \end{bmatrix}, \mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathrm{N}$

If trace $(A) = \text{trace}(B)$ and a, b, c are sides of ΔABC with $BC = a, CA = b$ and $AB = c$, then $\cos A$ is -

(A) $\frac{-79}{120}$

(B) $\frac{-89}{120}$

(C) $\frac{-33}{40}$

(D) $\frac{-31}{40}$

Q.8

If $\begin{vmatrix} \frac{\mathrm{a}}{\mathrm{b}} + \frac{\mathrm{a}}{\mathrm{c}} & \frac{\mathrm{b}}{\mathrm{c}} + \frac{\mathrm{b}}{\mathrm{a}} & 1 \\ \frac{\mathrm{b}}{\mathrm{a}} + \frac{\mathrm{b}}{\mathrm{c}} & \frac{\mathrm{c}}{\mathrm{a}} + \frac{\mathrm{c}}{\mathrm{b}} & 1 \\ \frac{\mathrm{c}}{\mathrm{a}} + \frac{\mathrm{c}}{\mathrm{b}} & \frac{\mathrm{a}}{\mathrm{b}} + \frac{\mathrm{a}}{\mathrm{c}} & 1 \end{vmatrix} = 0$

(A) $\frac{1}{\mathrm{a}} + \frac{1}{\mathrm{b}} + \frac{1}{\mathrm{c}} = 0$

(B) $a^2 + b^2 + c^2 = ab + bc + ac$

(C) $\mathrm{a} = \mathrm{b} \neq \mathrm{c}$

(D) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$

Q.9

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Let $\Delta(x)=\begin{vmatrix} 3 & 3x & 3x^2+2a^2 \\ 3x & 3x^2+2a^2 & 3x^3+6a^2x \\ 3x^2+2a^2 & 3x^3+6a^2x & 3x^4+12a^2x^2+2a^4 \end{vmatrix}$ then -

(A) $\Delta'(x)=0$

(B) Δ (x) is independent of x

(C) $\int_0^1 \Delta(x)dx=16a^6$

(D) $y=\Delta$ (x) is a straight line of infinite slope

Q.10

System of equation : $x+3y+2z=6; x+\lambda y+2z=7; x+3y+2z=\mu$ has

(A) unique solution if $\lambda=2, \mu\neq 6$

(B) infinitely many solution if $\lambda=4, \mu=6$

(C) no solution if $\lambda=5, \mu=7$

(D) no solution if $\lambda=3, \mu=5$

Q.11

Let $M=[m_{ij}]_{3\times 3}$ matrix where $m_{ij}=\begin{cases} 2\sin t & i=j \\ 1 & |j-i|=1 \\ 0 & \text{otherwise} \end{cases}$, then -

(A) M is a symmetric matrix.

(B) Square of the maximum value of $\det(M)$ is 16 .

(C) Square of the minimum value of $\det(M)$ is 16 .

(D) Maximum value of determinant of co-factor matrix is 16 .

Q.12

Passage

Let $\Delta(x)=\begin{vmatrix} 2x^3-3x^2 & 5x+7 & 2 \\ 4x^3-7x & 3x+2 & 1 \\ 7x^3-8x^2 & x-1 & 3 \end{vmatrix}=a_0+a_1x+\ldots\ldots\ldots+a_4x^4$

To evaluate a_i we differentiate $\Delta(x)$ i times w.r.t. x and put $x=0$ or divide $\Delta(x)$ by x^4 put $1/x=t$, differentiate $(4-$ i) time w.r.t. x and put $t=0$

Question:

a_1 equals -

(A) 0

(B) 61

(C) 161

(D) 191

Q.13

Passage

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Let $\Delta(x)=\begin{vmatrix} 2x^3-3x^2 & 5x+7 & 2 \\ 4x^3-7x & 3x+2 & 1 \\ 7x^3-8x^2 & x-1 & 3 \end{vmatrix}=a_0+a_1x+\ldots\ldots\ldots+a_4x^4$

To evaluate a_i we differentiate $\Delta(x)$ i times w.r.t. x and put $x=0$ or divide $\Delta(x)$ by x^4 put $1/x=t$, differentiate $(4-i)$ time w.r.t. x and put $t=0$

Question:

a $_3$ equals -

(A) -70

(B) -73

(C) -74

(D) 0

Q.14

Passage

Let $\Delta(x)=\begin{vmatrix} 2x^3-3x^2 & 5x+7 & 2 \\ 4x^3-7x & 3x+2 & 1 \\ 7x^3-8x^2 & x-1 & 3 \end{vmatrix}=a_0+a_1x+\ldots\ldots\ldots+a_4x^4$

To evaluate a_i we differentiate $\Delta(x)$ i times w.r.t. x and put $x=0$ or divide $\Delta(x)$ by x^4 put $1/x=t$, differentiate $(4-i)$ time w.r.t. x and put $t=0$

Question:

a_4 equals -

(A) 46

(B) -43

(C) 41

(D) 0

Q.15

Passage

Let A be a $m\times n$ matrix. If there exists a matrix L of type $n\times m$ such that $LA=I_n$, then L is called left inverse of A. Similarly, if there exists a matrix R of type $n\times m$ such that $AR=I_m$, then R is called right inverse of A.

For example to find right inverse of matrix.

$A=\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}$ we take $R=\begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}$ and solve $AR=I_3$ i.e.

$\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$x-u=1$

$y-v=0$

$z-w=0$

$x+u=0$

$y+v=1$

$z+w=0$

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$2x + 3u = 0$
$2y + 3v = 0$
$2z + 3w = 1$

At this system of equation is inconsistent, we say there is no right inverse for matrix A.

Question:

Which of the following matrices is not left inverse of matrix $\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}$

(A) $\begin{bmatrix} 1/2 & 1/2 & 0 \\ -1/2 & 1/2 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & -7 & 3 \\ -1/2 & 1/2 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} -1/2 & 1/2 & 0 \\ -1/2 & 1/2 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & 3 & -1 \\ -1/2 & 1/2 & 0 \end{bmatrix}$

Q.16

Passage

Let A be a $m \times n$ matrix. If there exists a matrix L of type $n \times m$ such that $LA = I_n$, then L is called left inverse of A. Similarly, if there exists a matrix R of type $\mathrm{n} \times \mathrm{m}$ such that $\mathrm{AR} = \mathrm{I_m}$, then R is called right inverse of A.

For example to find right inverse of matrix.

$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}$ we take $R = \begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}$ and solve $AR = I_3$ i.e.

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \mathrm{x} & \mathrm{y} & \mathrm{z} \\ \mathrm{u} & \mathrm{v} & \mathrm{w} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{lll} x - u = 1 & y - v = 0 & z - w = 0 \\ x + u = 0 & y + v = 1 & z + w = 0 \\ 2x + 3u = 0 & 2y + 3v = 0 & 2z + 3w = 1 \end{array}$$

At this system of equation is inconsistent, we say there is no right inverse for matrix A.

The number of right inverses for the matrix $\begin{bmatrix} 1 & -1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$

(A) 0

(B) 1

(C) 2

(D) infinite

Q.17

Passage

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Let A be a $m \times n$ matrix. If there exists a matrix L of type $n \times m$ such that $LA = I_n$, then L is called left inverse of A. Similarly, if there exists a matrix R of type $\mathrm{n} \times \mathrm{m}$ such that $\mathrm{AR} = \mathrm{I_m}$, then R is called right inverse of A.

For example to find right inverse of matrix.

$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}$ we take $R = \begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}$ and solve $AR = I_3$ i.e.

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \mathrm{x} & \mathrm{y} & \mathrm{z} \\ \mathrm{u} & \mathrm{v} & \mathrm{w} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{lll} x - u = 1 & y - v = 0 & z - w = 0 \\ x + u = 0 & y + v = 1 & z + w = 0 \\ 2x + 3u = 0 & 2y + 3v = 0 & 2z + 3w = 1 \end{array}$$

At this system of equation is inconsistent, we say there is no right inverse for matrix A.

Question:

For which of the following matrices number of left inverses is greater than the number of right inverses -

(A) $\begin{bmatrix} 1 & 2 & 4 \\ -3 & 2 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 4 \\ 2 & -3 \\ 5 & 4 \end{bmatrix}$

(D) $\begin{bmatrix} 3 & 3 \\ 1 & 1 \\ 4 & 4 \end{bmatrix}$

Q.18

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Match the column -

Column I	Column II
(A) If $A=\begin{bmatrix}2 & -1\\ 3 & 2\end{bmatrix}, B=\begin{bmatrix}0 & 4\\ -1 & 7\end{bmatrix}$, then values which are lesser than $[\,\lvert 3A^2-2B\rvert-700]/9$	(p) 0
(B) $A=\begin{bmatrix}2 & 3\\ 5 & -2\end{bmatrix}$ and $A^{-1}=\frac{1}{k}A$, then the values which are greater than $(k-15)/4$	(q) 1
(C) If $\begin{vmatrix}0 & ab^2 & ac^2\\ a^2b & 0 & bc^2\\ a^2c & cb^2 & 0\end{vmatrix}=2a^pb^qc^r$, then the values which are equal to or greater than $(p+q+r)/3$	(r) 2
(D) If $A=\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}$ and $A^n=O$, then the minimum value of n is	(s) 3
	(t) 4

(A) A-p,q,s ; B-r,s,t ; C-s,q ; D-p

(B) A-p,q,t ; B-r,s,p ; C-s,p ; D-p

(C) A-p,q,r ; B-r,s,t ; C-s,t ; D-r

(D) A-p,q,r ; B-r,s,p ; C-q,r ; D-r

Q.19

If A is a square matrix of order 3 such that $|A|=2$ then $\left|\left(\text{adj}A^{-1}\right)^{-1}\right|$ is -

Q.20

Let $D=\begin{vmatrix}a & a^2 & 0\\ 1 & 2a+b & (a+b)^2\\ 0 & 1 & 2a+3b\end{vmatrix}$, if $a+x_1b$ and $a+x_2b$ is a factor of D then find the value of x_1+x_2, where $x_1, x_2 \neq 0$

Answers & Solutions

Q.1 Answer:

24

Solution:

$$\begin{vmatrix}y & z & y\\ 0 & z+x & x\\ y & x & x+y\end{vmatrix}+\begin{vmatrix}z & z & y\\ z & z+x & x\\ 0 & x & x+y\end{vmatrix}=\begin{vmatrix}y & z & y\\ 0 & z+x & x\\ 0 & x-z & x\end{vmatrix}+\begin{vmatrix}z & z & y\\ 0 & x & x-y\\ 0 & x & x+y\end{vmatrix}=4xyz$$

$D=8 \Rightarrow |xyz|=2 \Rightarrow |x||y||z|=|\pm 2||\pm 1||\pm 1|.$

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Number of solutions $= 24$

Q.2 Answer:

-I

Solution:

$$f(\alpha)f(\beta) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & \cos\alpha\sin\beta + \sin\alpha\cos\beta \\ -\sin\alpha\cos\beta - \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{bmatrix} = \begin{bmatrix} \cos(\alpha+\beta) & \sin(\alpha+\beta) \\ -\sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix}$$

Similarly $f(\alpha)f(\beta)f(\gamma) = \begin{bmatrix} \cos(\alpha+\beta+\gamma) & \sin(\alpha+\beta+\gamma) \\ -\sin(\alpha+\beta+\gamma) & \cos(\alpha+\beta+\gamma) \end{bmatrix} = \begin{bmatrix} \cos\pi & \sin\pi \\ -\sin\pi & \cos\pi \end{bmatrix}$ **as** $\alpha + \beta + \gamma = \pi$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -\mathrm{I}$$

Q.3 Answer:

-4

Solution:

Let $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$; $C_2 \to C_2 - \frac{a_{12}}{a_{11}}C_1; C_3 \to C_3 - \frac{a_{13}}{a_{11}}C_1$

$$\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & \left(a_{22} - \frac{a_{12}}{a_{11}} \times a_{21}\right) & \left(a_{23} - \frac{a_{13}}{a_{11}} \times a_{21}\right) \\ a_{31} & \left(a_{32} - \frac{a_{12}}{a_{11}} \times a_{31}\right) & \left(a_{32} - \frac{a_{13}}{a_{11}} \times a_{31}\right) \end{vmatrix}$$

so minimum value $= -4$

Q.4 Answer:

0

Solution:

$$\frac{d^n[f(x)]}{dx^n \text{ i.e.} f^n(x) =} \begin{vmatrix} \frac{d^n}{dx^n}(x^n) & \frac{d^n}{dx^n}(\sin x) & \frac{d^n}{dx^n}(\cos x) \\ n! & \sin\left(\frac{n\pi}{2}\right) & \cos\left(\frac{n\pi}{2}\right) \\ a & a^2 & a^3 \end{vmatrix}$$

[since elements in second and third rows are constants]

Let $y = x^n$, **then** $y_1 = \frac{dy}{dx} = nx^{n-1}, y_2 = \frac{d^2y}{dx^2} = n(n-1)x^{n-1}, \ldots\ldots$

$y_n = n(n-1)(n-2)\ldots..1 \cdot x^{n-n} = n!$

Again let $y = \sin x$, **then** $y_1 = \cos x = \sin\left(\frac{\pi}{2} + x\right)$

$y_2 = \cos\left(\frac{\pi}{2} + x\right) = \sin\left\{\frac{\pi}{2} + \left(\frac{\pi}{2} + x\right)\right\} = \sin\left(2 \cdot \frac{\pi}{2} + x\right)$

Proceeding in this way $y_n = \sin\left(n \cdot \frac{\pi}{2} + x\right)$

Similarly, if $y = \cos x$, **then** $y_1 = -\sin x = \cos\left(\frac{\pi}{2} + x\right)$

$y_2 = -\sin\left(\frac{\pi}{2} + x\right) = \cos\left\{\frac{\pi}{2} + \left(\frac{\pi}{2} + x\right)\right\} = \cos\left(2 \cdot \frac{\pi}{2} + x\right)$

$\therefore \quad y_n = \cos\left(n \cdot \frac{\pi}{2} + x\right)$

Thus, $\frac{d^n}{dx^n}(x^n) = n!, \frac{d^n}{dx^n}(\sin x) = \sin\left(x + \frac{n\pi}{2}\right)$ **and** $\frac{d^n}{dx^n}(\cos x) = \cos\left(x + \frac{n\pi}{2}\right)$

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From (1), $f^n(x) = \begin{vmatrix} n! & \sin\left(x+\frac{n\pi}{2}\right) & \cos\left(x+\frac{n}{2}\right) \\ n! & \sin\left(\frac{n\pi}{2}\right) & \cos\left(\frac{n\pi}{2}\right) \\ a & a^2 & a^3 \end{vmatrix}$

$\therefore f^n(0) = \begin{vmatrix} n! & \sin\left(0+\frac{n\pi}{2}\right) & \cos\left(0+\frac{n\pi}{2}\right) \\ n! & \sin\left(\frac{n\pi}{2}\right) & \cos\left(\frac{n\pi}{2}\right) \\ a & a^2 & a^3 \end{vmatrix} = 0$ [R_1 **and** R_2 **are identical**]

Q.5 Answer:

$$I + A = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

Solution:

If $\alpha = \pi/2$ **then** $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$$I + A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$I - A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

(a) R.H.S. $= \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \neq I + A$

(c) R.H.S. $= \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = I + A = L.H.S.$

Q.6 Answer:

$A^2 - B^2$

Solution:

$$(A+B) = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}, (A-B) = \begin{bmatrix} i & i \\ i & i \end{bmatrix}$$

$$(A+B)(A-B) = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}\begin{bmatrix} i & i \\ i & i \end{bmatrix} = \begin{bmatrix} i^2 - i^2 & i^2 - i^2 \\ -i^2 + i^2 & -i^2 + i^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} i^2 & 0 \\ 0 & i^2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}\begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = \begin{bmatrix} i^2 & 0 \\ 0 & i^2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} 0 & -i^2 \\ -i^2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}\begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = \begin{bmatrix} 0 & -i^2 \\ -i^2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Hence, $A^2 - B^2 = 0$

Q.7 Answer:

$\frac{-89}{120}$

Solution:

$5a^2 + 2bc + 8\,b^2 - 10ac + 25c^2 = a^2 + 6ab - b^2 + 17bc$

$4a^2 + 9\,b^2 + 25c^2 = (2a)(3\,b) + 15bc + 10ac$

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$\Rightarrow (2a)^2+(3b)^2+(5c)^2=(2a)(3b)+(3b)(5c)+(5c)(2a)$

$\Rightarrow \quad 2a=3b=5c$

$\Rightarrow \quad \frac{a}{15}=\frac{b}{10}=\frac{c}{6}=\lambda \text{ (say)}$

$$\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{100\lambda^2+36\lambda^2-225\lambda^2}{2(10\lambda)(6\lambda)}=\frac{-89\lambda^2}{120\lambda^2}=\frac{-89}{120}$$

Q.8 Answer:

$a^2+b^2+c^2=ab+bc+ac$

Solution:

$C_1 \to C_1+C_3, C_2 \to C_2+C_3$

$$\Delta=\begin{vmatrix} a\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) & b\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) & 1 \\ b\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) & c\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) & 1 \\ c\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) & a\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) & 1 \end{vmatrix}=\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)^2\begin{vmatrix} a & b & 1 \\ b & c & 1 \\ c & a & 1 \end{vmatrix}$$

Since a, b, c are positive real numbers the first bracket cannot be zero. $\Rightarrow \quad a^2+b^2+c^2-ab-$

$\Rightarrow \quad a=b=c$

Q.9 Answer:

$\Delta'(x)=0$

Solution:

Applying $C_3 \to C_3-xC_2, C_2 \to C_2-xC_1$

$$\text{we obtain } \Delta(x)=\begin{vmatrix} 3 & 0 & 2a^2 \\ 3x & 2a^2 & 4a^2x \\ 3x^2+2a^2 & 4a^2x & 6a^2x^2+2a^4 \end{vmatrix}$$

$$\text{Apply } C_3 \to C_3-xC_2, \text{ we get } \Delta(x)=4a^4\begin{vmatrix} 3 & 0 & 1 \\ 3x & 1 & x \\ 3x^2+2a^2 & 2x & x^2+2a^2 \end{vmatrix}$$

$$\text{Apply } C_1 \to C_1-3C_3 \text{ to get } \Delta(x)=4a^4\begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & x \\ -4a^2 & 2x & x^2+2a^2 \end{vmatrix}=16a^6$$

$\therefore \quad \Delta'(x)=0, \Delta x$ is independent of x,

$\int_0^1 \Delta(x)dx=16a^6$, and $y=16a^6$ is a straight line of zero slope.

Q.10 Answer:

infinitely many solution if $\lambda=4, \mu=6$

Solution:

$x+3y+2z=6$

$x+\lambda y+2z=7$

$x+3y+2z=\mu$

(A) If $\lambda=2$, then $D=0$, therefore unique solution is not possible.

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(B) If $\lambda = 4, \mu = 6$

$x + 3y = 6 - 2z$

$x + 4y = 7 - 2z$

$\therefore \quad y = 1$ and $x = 3 - 2z$

Substituting in eq. (3)

$3 - 2z + 3 + 2z = 6$ **is satisfied**

$\therefore$ **infinite solutions**

(C) If $\lambda = 5, \mu = 7$

Consider eq. (2) and (3)

$x + 5y = 7 - 2z$

$x + 3y = 7 - 2z$

$y = 0, x = 7 - 2z$ **are solution**

sub. in (1)

$7 - 2z + 2z = 6$ **does not satisfy**

$\therefore$ no solution

(D) If If $\lambda = 3, \mu = 5$

then eq. (1) and (2) have no solution

$\therefore$ no solution.

Q.11 Answer:

M is a symmetric matrix.

Solution:

$$M = \begin{bmatrix} 2\sin t & 1 & 0 \\ 1 & 2\sin t & 1 \\ 0 & 1 & 2\sin t \end{bmatrix} = M^T \Rightarrow M \text{ is symmetric matrix}$$

$|M| = 8\sin^3 t - 4\sin t$

Let $\sin t = \alpha \Rightarrow |M| = 8\alpha^3 - 4\alpha \quad -1 \le \alpha \le 1$

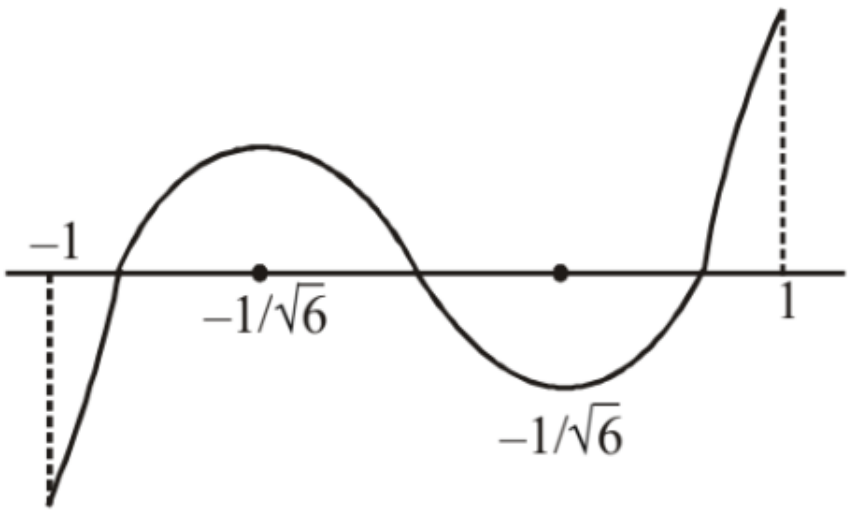
$\Rightarrow f(\alpha) = 8\alpha^3 - 4\alpha \Rightarrow f'(\alpha) = 24\alpha^2 - 4 \Rightarrow f'(\alpha) = 0$

$\Rightarrow \alpha = \pm\frac{1}{\sqrt{6}}$

Hence maximum and minimum value occur at $\alpha = 1$ **and** $\alpha = -1$ **respectively.**

$\Rightarrow \quad |M|_{max} = 4$ **and** $|M|_{min} = -4$

$\Rightarrow |M|^2 \max / \min = 16$



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Det (co. factor matrix) $= |\mathrm{M}|^{\mathrm{n}-1} = |\mathrm{M}|^2 = (4)^2 = 16$

Q.12 Answer:

161

Solution:

$$\Delta'(x) = \begin{vmatrix} 6x^2-6x & 5x+7 & 2 \\ 12x^2-7 & 3x+2 & 1 \\ 21x^2-16x & x-1 & 3 \end{vmatrix} + \begin{vmatrix} 2x^3-3x^2 & 5 & 2 \\ 4x^3-7x & 3 & 1 \\ 7x^3-8x^2 & 1 & 3 \end{vmatrix}$$

$a_1 = \Delta'(0) = 161$

Q.13 Answer:

-73

Solution:

Put $1/\mathrm{x} = \mathrm{t}$ in $\Delta(\mathrm{x})/\mathrm{x}^4$ and write $\Delta_1(\mathrm{t}) = \begin{vmatrix} 2-3\mathrm{t} & 5+7\mathrm{t} & 2 \\ 4-7\mathrm{t}^2 & 3+2\mathrm{t} & 1 \\ 7-8\mathrm{t} & 1-\mathrm{t} & 3 \end{vmatrix}$

$$\Delta_1'(\mathrm{t}) = \begin{vmatrix} -3 & 5+7\mathrm{t} & 2 \\ -14\mathrm{t} & 3+2\mathrm{t} & 1 \\ -8 & 1-\mathrm{t} & 3 \end{vmatrix} + \begin{vmatrix} 2-3\mathrm{t} & 7 & 2 \\ 4-7\mathrm{t}^2 & 2 & 1 \\ 7-8\mathrm{t} & -1 & 3 \end{vmatrix}$$

$\mathrm{a}_3 = \Delta_1'(0) = -73$

Q.14 Answer:

-43

Solution:

$$\frac{\Delta(\mathrm{x})}{\mathrm{x}^4} = \begin{vmatrix} 2-3/\mathrm{x} & 5+7/\mathrm{x} & 2 \\ 4-7/\mathrm{x}^2 & 3+2/\mathrm{x} & 1 \\ 7-8/\mathrm{x} & 1-1/\mathrm{x} & 3 \end{vmatrix}$$

Taking limit as $x \to \infty$, we get $a_4 = \begin{vmatrix} 2 & 5 & 2 \\ 4 & 3 & 1 \\ 7 & 1 & 3 \end{vmatrix} = -43$

Q.15 Answer:

$$\begin{bmatrix} -1/2 & 1/2 & 0 \\ -1/2 & 1/2 & 0 \end{bmatrix}$$

Solution:

As second row of all the options is same, we look at the elements of the first row.

Let left inverse be $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$**. Then**

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore,

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$a+b+2c=1$

$-a+b+3c=0$

$\Rightarrow b=\dfrac{1-5c}{2}$ and $a=\dfrac{1+c}{2}$

Thus, matrices in options $(A),(B)$ and (D) are the inverses and matrix in option (C) is not the left inverse.

Q.16 Answer:

infinite

Solution:

A is a matrix of order $m \times n$.

If there exists a matrix R of type $n \times m$ such that $AR = I_m$, then R is called right inverse of A.

Let $R=\begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & -1 & 1 \end{bmatrix}\begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$a-b+2c=1, d-e+2f=0; 2a-b+c=0, 2d-e+f=1$

Three variables in two equations, hence above system has infinite solutions.

$\therefore$ Infinite right inverses are possible.

Hence, option D.

Q.17 Answer:

$$\begin{bmatrix} 1 & 4 \\ 2 & -3 \\ 5 & 4 \end{bmatrix}$$

Solution:

By observation there can't be any left inverse for (B) and (D) so we will check for (A) and (C) only.

For (A) let left inverse be $\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$**, then** $\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ -3 & 2 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Now $a-3b=1, 2a+2b=0$ and $4a+b=0$ which is not possible

For $(C):\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix} 1 & 4 \\ 2 & -3 \\ 5 & 4 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \quad a+2b+5c=1, 4a-3b+4c=0, d+2e+5f=0,$

$4d-3e+4f=1$

$\therefore$ there are infinite number of left inverses.

$$\begin{bmatrix} 1 & 4 \\ 2 & -3 \\ 5 & 4 \end{bmatrix}\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow a+4d=1, 2a-3d=0$ and $5a+4d=0$

which is not possible

$\therefore$ There is no right inverse.

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Q.18 Answer:

A-p,q,r ; B-r,s,t ; C-s,t ; D-r

Solution:

(A) - p,q,r, (B) - r,s,t, (C) - s,t, (D) - r

(A) $A^2=\begin{bmatrix}2&-1\\3&2\end{bmatrix}\begin{bmatrix}2&-1\\3&2\end{bmatrix}=\begin{bmatrix}1&-4\\12&1\end{bmatrix}$

$3A^2-2B=\begin{bmatrix}3&-12\\36&3\end{bmatrix}-\begin{bmatrix}0&8\\-2&14\end{bmatrix}=\begin{bmatrix}3&-20\\38&-11\end{bmatrix}$

$\left|3A^2-2B\right|=33+760=727$

$\left[\left|3\,\mathrm{A}^2-2\,\mathrm{B}\right|-700\right]/9=3$

(B) $A=\begin{bmatrix}2&3\\5&-2\end{bmatrix}$

$\mathrm{A}^{-1}=\frac{1}{-19}\begin{bmatrix}-2&-3\\-5&2\end{bmatrix}=\frac{1}{19}\begin{bmatrix}2&3\\5&-2\end{bmatrix}=\frac{1}{19}\,\mathrm{A}$

$\therefore\quad \mathrm{k}=19$

$\therefore(\mathrm{k}-15)/4=1$

(C) $\begin{vmatrix}0&ab^2&ac^2\\a^2b&0&bc^2\\a^2c&cb^2&0\end{vmatrix}=abc\begin{vmatrix}0&b^2&c^2\\a^2&0&c^2\\a^2&b^2&0\end{vmatrix}=abc\left[-a^2\left(0-b^2c^2\right)+a^2\left(b^2c^2-0\right)\right]=2a^3b^3c^3$

$p+q+r=9$

$(p+q+r)/3=3$

(D) $A^2=A\cdot A=\begin{bmatrix}ab&b^2\\-a^2&-ab\end{bmatrix}\begin{bmatrix}ab&b^2\\-a^2&-ab\end{bmatrix}=\begin{bmatrix}a^2b^2-a^2b^2&ab^3-ab^3\\-a^3b+a^3b&-a^2b^2+a^2b^2\end{bmatrix}=0$

$\Rightarrow \mathrm{A}^3=\mathrm{A}\cdot\mathrm{A}^2=0$ **and** $\mathrm{A}^{\mathrm{n}}=0$**, for all** $\mathrm{n}\geq 2$**.**

Q.19 Answer:

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Solution:

$\left|\operatorname{adj}A^{-1}\right|=\left|A^{-1}\right|^2=\frac{1}{|A|^2}$

$\left|\left(\operatorname{adj}A^{-1}\right)^{-1}\right|=\frac{1}{\left|\operatorname{adj}A^{-1}\right|}=|A|^2=2^2=4$

Q.20 Answer:

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Solution:

On applying C_2-aC_1 **we easily get** $D=a(a+b)(a+2b)$

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